General Mathematics — Exercise Sheet 29

Numerical and signed series

Variant: Statements

Exercise 1 (Choose the test and retain its hypotheses). Classify each series as absolutely convergent, conditionally convergent, or divergent. All sums start at $n = 1$ except part (c), which starts at $n = 2$. The parameter p in part (c) ranges over $\mathbb{R}$.

(a) $\displaystyle\sum_{n=1}^{\infty} (\sqrt{n^2+3} - n)$.

(b) $\displaystyle\sum_{n=1}^{\infty} (-1)^n \frac{4n+1}{n^3+2}$.

(c) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^p}$.

(d) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n+\sqrt{n}}$. Also find the least number $N \geq 1$ of terms for which Lecture 29's alternating bound guarantees error at most $1/100$.

(e) $\displaystyle\sum_{n=1}^{\infty} \left(\frac{n+1}{3n+2}\right)^{2n}$.

(f) $\displaystyle\sum_{n=1}^{\infty} \frac{n!}{n^2 2^n}$.

Use Lecture 29's tests; the logarithmic improper-integral criterion may be recalled from Sheet 27, or re-derived by finite substitution $u = \ln x$.

Exercise 2 (The root test, with zero terms allowed). Prove the root test stated in Lecture 29: if a real sequence satisfies $|a_n|^{1/n} \to \ell \in [0, \infty]$, then its series converges absolutely when $\ell < 1$, and diverges when $\ell > 1$. Explain why no eventual nonzero assumption is required. Show that $\ell = 1$ gives no conclusion by exhibiting one convergent and one divergent series with that root limit. Use geometric comparison and the term test of Lecture 29, and $n^{1/n} \to 1$ from Lecture 4.

Exercise 3 (Alternation without monotonicity). For integers $k \geq 1$, define $b_{2k-1} = 1/k$ and $b_{2k} = 1/(2k)$. Prove $b_n \to 0$, but show that the alternating series $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ diverges. Identify the failed hypothesis in the alternating-series theorem.

Exercise 4 (A second cancellation in a finite sum). For $N \geq 2$, compute the partial sum $T_N = \sum_{n=2}^{N} 1/(n(n^2-1))$, then determine the infinite sum and its exact remainder after index N. Use the telescoping method of Lecture 29, retaining all boundary terms.

Exercise 5 (⋆ Rearranging real series). A rearrangement is specified by a bijection $\sigma : \mathbb{N} \to \mathbb{N}$, as in Lecture 29.

(a) Prove that if $\sum a_n$ converges absolutely to S, every rearranged series $\sum a_{\sigma(n)}$ also converges to S.

(b) Suppose instead that $\sum a_n$ converges conditionally. Prove that for every prescribed $T \in \mathbb{R}$ there is a rearrangement converging to T. Include zero terms in the bijection if they occur.

Use Lecture 29's partial-sum definition, positive and negative parts, comparison and term test, together with Lecture 4's monotone convergence, squeeze theorem and subsequence law. Nothing later in the course depends on part (b).